Radian measure

1

a State the multiplier to convert an angle from degrees to radians.

b State the multiplier to convert an angle from radians to degrees.

2 Convert the following to radians, correct to two decimal places where applicable:

a 360° **b** 180° **c** 90° **d** 225°

e -300° **f** 112° **g** 29° **h** 161.17°

3 Convert the following to degrees:

a $\frac{\pi}{3}$ **b** $\frac{\pi}{6}$ **c** $\frac{\pi}{4}$ **d** $\frac{3\pi}{2}$

e $\frac{5\pi}{4}$ **f** $-\frac{2\pi}{3}$ **g** $-\frac{5\pi}{3}$ **h** 4.2 radians

4 Determine whether the following statements are true or false. If it is false, correct the statement.

a If $180^\circ = \pi$ radians, then 90° must be equal to $\frac{\pi}{2}$ radians.

b 60° must be equal to $\frac{\pi}{6}$ radians, because it is $\frac{1}{6}$ of 360°.

c 30° must be equal to $\frac{\pi}{3}$ radians, because it is half of 60°.

d 210° must be equal to $\frac{7\pi}{6}$ radians, because it is seven groups of 30°.

e 45° must be equal to $\frac{\pi}{4}$ radians, because it is half of 90°.

f 225° must be equal to $\frac{7\pi}{4}$ radians, because it is seven groups of 45°.

Coterminal angles

5 Consider the angle $\frac{5\pi}{8}$.

a State the measure of the coterminal angle to $\frac{5\pi}{8}$ that is between 2π and 4π.

b State the measure of the coterminal angle to $\frac{5\pi}{8}$ that is between 0 and -2π.

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6 Consider the angle $\frac{9\pi}{8}$.

a State the measure of the coterminal angle to $\frac{9\pi}{8}$ that is between 2π and 4π.

b State the measure of the coterminal angle to $\frac{9\pi}{8}$ that is between 0 and -2π.

7 Consider the angle $-\frac{2\pi}{3}$.

a State the measure of the coterminal angle to $-\frac{2\pi}{3}$ that is between 0 and 2π.

b State the measure of the coterminal angle to $-\frac{2\pi}{3}$ that is between -2π and -4π.

8 Consider the angle $\frac{4\pi}{3}$.

a In which quadrant is the terminal side of this angle?

b Determine the measure of the acute angle between the terminal side of $\frac{4\pi}{3}$ and the x-axis.

c Now, what acute angle in the first quadrant can $\frac{4\pi}{3}$ be related to?

9 Consider the angle $\frac{8\pi}{7}$.

a In which quadrant is the terminal side of this angle?

b Determine the measure of the acute angle between the terminal side of $\frac{8\pi}{7}$ and the x-axis.

c Now, what acute angle in the first quadrant can $\frac{8\pi}{7}$ be related to?

10 Consider an angle of 94°.

a Find the measure of the smallest positive angle that is coterminal with 94°.

b Find the first negative angle measure that is coterminal with 94°.

c State which quadrant does 94° lie in.

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11 Consider an angle of -58°.

a Find the measure of the smallest positive angle that is coterminal with -58°.

b Find the measure of the smallest negative angle that is coterminal with -58°.

c State which quadrant -58° lies in.

12 Find the measure of the smallest positive angle that is coterminal with each of the following angles:

a 489° **b** -331°

13 Find the measure of the smallest positive angle that is coterminal with each of the following angles, leaving your answers in radians:

a $\frac{9\pi}{4}$ **b** $-\frac{11\pi}{3}$

14 For each of the following, state the expression in terms of n, where n represents any integer, that generates all angles coterminal with the given angle:

a $240^\circ + \square$ **b** $-90^\circ + \square$

15 For each of the following, state the expression in terms of n, where n represents any integer, that generates all angles coterminal with the given angle:

a $\frac{\pi}{2} + \square$ **b** $-\frac{\pi}{2} + \square$

Reference angles

16 Find the measure of the reference angle for each of the following angles:

a 111° **b** 197° **c** 275°

17 For each of the following angles:

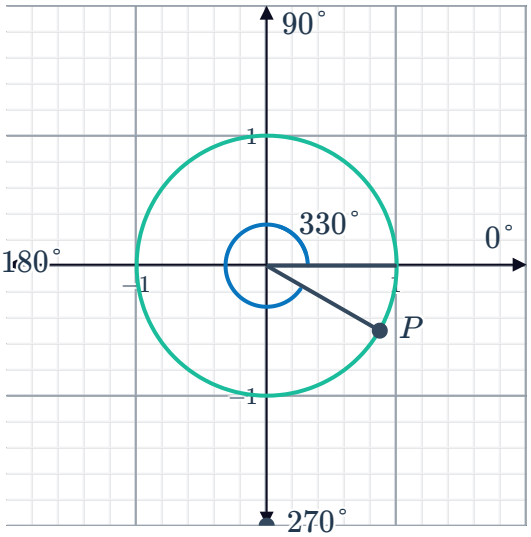
i Find the measure of the coterminal angle that is between 0° and 360°.

ii Find the measure of the reference angle.

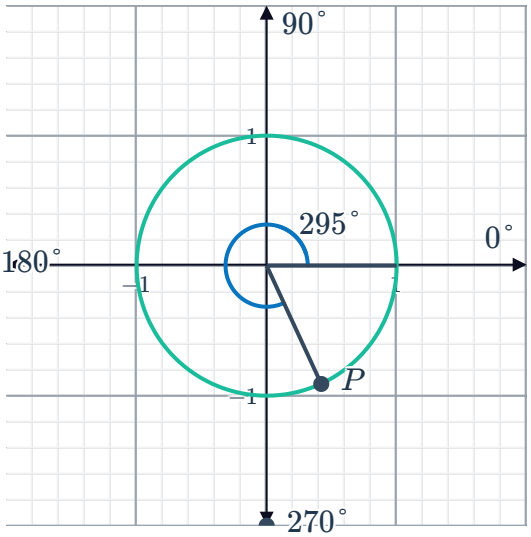
a -167° **b** 519°

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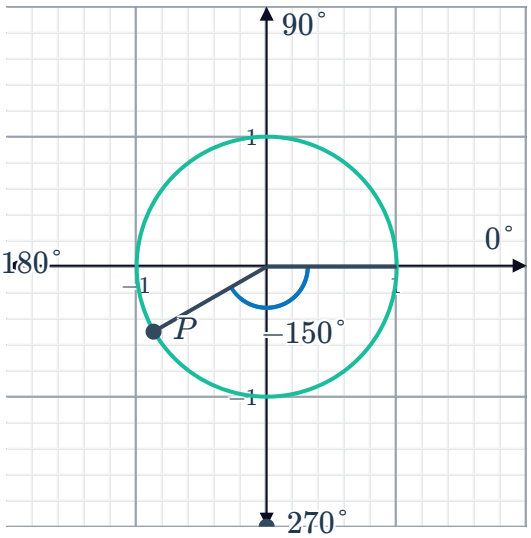
18 Point P on the unit circle shows a rotation of 330°.
Find the measure of the related acute angle in the first quadrant.



19 Point P on the unit circle shows a rotation of 295°.
Find the measure of the related acute angle in the first quadrant.



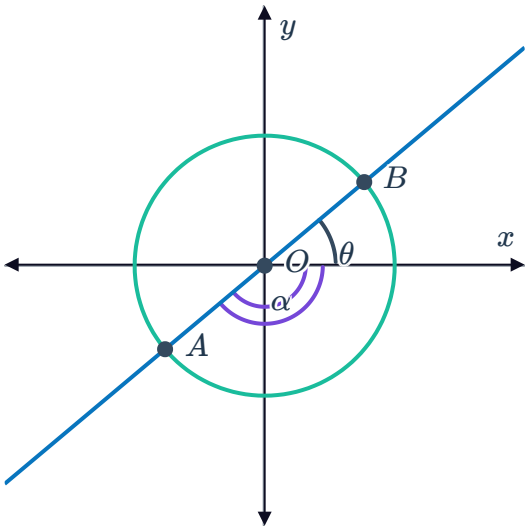
20 Point P on the unit circle shows a rotation of -150°.
Find the measure of the related acute angle in the first quadrant.



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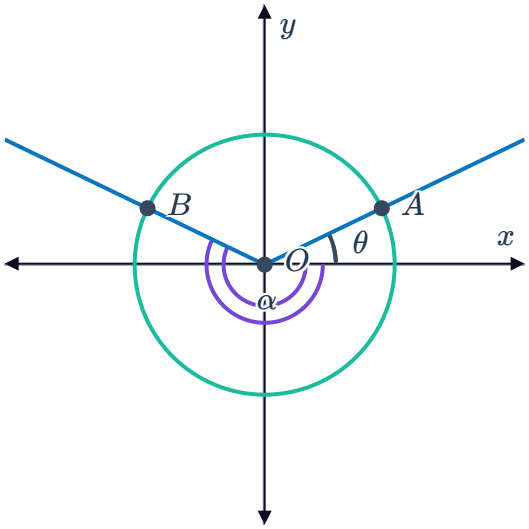
21 The graph shows a line through two points on the unit circle, A and B, and the origin, O. The line segment OB forms an angle of $\theta = \frac{\pi}{13}$ with the positive x-axis. The angle between the line segment OA and the positive x-axis is α.
Find the value of α when:

a $0 \leq \alpha < 2\pi$ **b** $-2\pi < \alpha \leq 0$



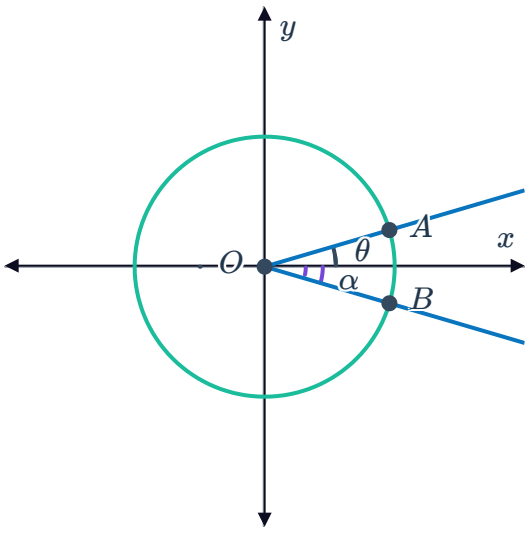
22 Points A and B are two points on the unit circle centered at the origin, O. The line segment OA forms an angle of $\theta = \frac{\pi}{7}$ with the positive x-axis, and the line passing through A and B is parallel to the x-axis. The angle between the line segment OB and the positive x-axis is α.
Find the value of α when:

a $0 \leq \alpha < 2\pi$ **b** $-2\pi < \alpha \leq 0$



23 Points A and B are two points on the unit circle centered at the origin, O. The line segment OA forms an angle of $\theta = \frac{\pi}{11}$ with the positive x-axis, and the line passing through A and B is parallel to the y-axis. The angle between the line segment OB and the positive x-axis is α.
Find the value of α when:

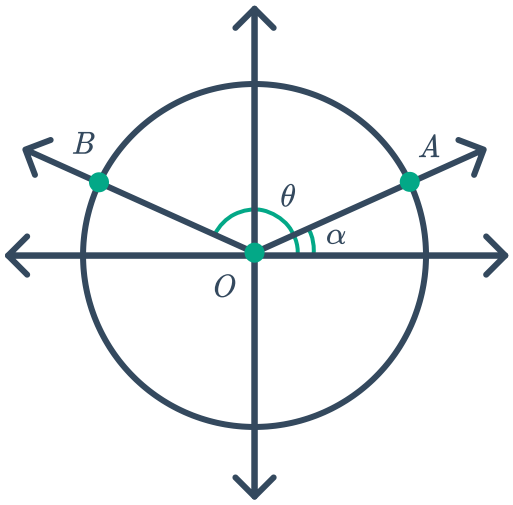
a $0 \leq \alpha < 2\pi$ **b** $-2\pi < \alpha \leq 0$



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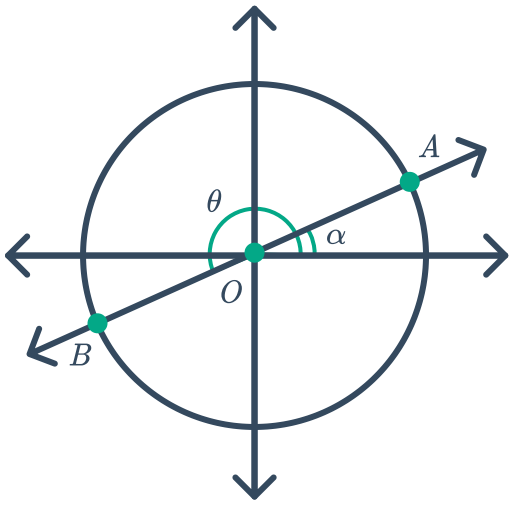
24 Consider the unit circle shown where points A and B have the same y-coordinates.

Suppose that $\theta = \dfrac{10\pi}{11}$. State the measure of the reference angle, α.



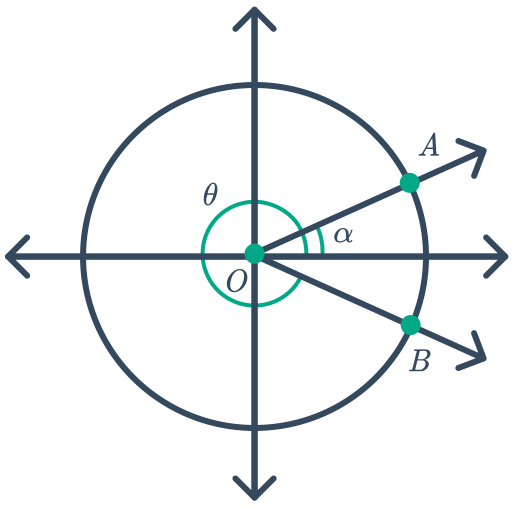
25 Consider the unit circle shown where the line through A and B passes through the origin, O.

Suppose that $\theta = \dfrac{8\pi}{7}$. State the size of the reference angle, α.



26 Consider the unit circle shown where points A and B have the same x-coordinate.

Suppose that $\theta = \dfrac{9\pi}{5}$. State the measure of the reference angle, α.



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27 Find the measure of the reference angle for $\frac{2\pi}{3}$.

28

a Find the measure of the coterminal angle to $-\frac{8\pi}{9}$ that is between 0 and 2π.

b Now, find the measure of the reference angle for $-\frac{8\pi}{9}$.

29 Consider the acute reference angle with measure $\frac{\pi}{17}$.
State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:

a $\sin\theta = \sin\frac{\pi}{17}$

b $\sin\theta = -\sin\frac{\pi}{17}$

c $\cos\theta = \cos\frac{\pi}{17}$

d $\cos\theta = -\cos\frac{\pi}{17}$

e $\tan\theta = \tan\frac{\pi}{17}$

f $\tan\theta = -\tan\frac{\pi}{17}$

30 Consider the angle with measure $\frac{9\pi}{13}$.

a State the measure of the acute reference angle for $\frac{9\pi}{13}$.

b State all values of α over the domain $0 \leq \alpha \leq 2\pi$ for which:

i $\sin\alpha = \sin\frac{9\pi}{13}$

ii $\cos\alpha = \cos\frac{9\pi}{13}$

iii $\tan\alpha = \tan\frac{9\pi}{13}$

31 Consider the angle $\frac{18\pi}{17}$.

a State the measure of the reference angle for $\frac{18\pi}{17}$.

b State all values of α over the domain $0 \leq \alpha \leq 2\pi$ for which:

i $\sin\alpha = \sin\frac{18\pi}{17}$

ii $\cos\alpha = \cos\frac{18\pi}{17}$

iii $\tan\alpha = \tan\frac{18\pi}{17}$

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32 Consider the angle $\frac{32\pi}{17}$.

a State the measure of the reference angle for $\frac{32\pi}{17}$.

b State all values of α over the domain $0 \leq \alpha \leq 2\pi$ for which:

i $\sin\alpha = \sin\frac{32\pi}{17}$

ii $\cos\alpha = \cos\frac{32\pi}{17}$

iii $\tan\alpha = \tan\frac{32\pi}{17}$

33 Consider the angle $\frac{9\pi}{17}$.

a State the measure of the reference angle for $\frac{9\pi}{17}$.

b State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:

i $\sin\theta = -\sin\frac{9\pi}{17}$

ii $\cos\theta = -\cos\frac{9\pi}{17}$

iii $\tan\theta = -\tan\frac{9\pi}{17}$

34 Consider the angle $\frac{19\pi}{17}$.

a State the measure of the reference angle for $\frac{19\pi}{17}$.

b State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:

i $\sin\theta = -\sin\frac{19\pi}{17}$

ii $\cos\theta = -\cos\frac{19\pi}{17}$

iii $\tan\theta = -\tan\frac{19\pi}{17}$

35 Consider the angle $\frac{21\pi}{13}$.

a State the measure of the reference angle for $\frac{21\pi}{13}$.

b State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:

i $\sin\theta = -\sin\frac{21\pi}{13}$

ii $\cos\theta = -\cos\frac{21\pi}{13}$

iii $\tan\theta = -\tan\frac{21\pi}{13}$

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